

corrigé zakito

Énoncé

Autocorrection A

$$1 \in M^\times, \quad a, b \in M^\times \Rightarrow (ab)(b^{-1}a^{-1}) = (b^{-1}a^{-1})(ab) = 1.$$

$$a^{-1} \in M^\times, \quad (a^{-1})^{-1} = a.$$

L'associativité est celle de M .

Exercice 1

1

e neutre de $*$, u neutre de $\otimes$.

$$e = (e \otimes u) * (u \otimes e) = (e * u) \otimes (u * e) = u \otimes u = u.$$

2

$$a * b = (a \otimes e) * (e \otimes b) = (a * e) \otimes (e * b) = a \otimes b.$$

3

$$a * b = (e \otimes a) * (b \otimes e) = (e * b) \otimes (a * e) = b \otimes a = b * a.$$

$$(a * b) * c = (a \otimes b) * (e \otimes c) = (a * e) \otimes (b * c) = a * (b * c).$$

Exercice 2

$$a, b \in A, \quad y, z \in E,$$

$$(a * b) * (y * z) = a * (b * (y * z)) = a * ((b * y) * z) = (a * (b * y)) * z = ((a * b) * y) * z.$$

$$a * b \in A, \quad a, b, c \in A \Rightarrow a * (b * c) = (a * b) * c.$$

Exercice 3

L'assertion est fausse pour un magma quelconque.

$$E = \{0, 1\}, \quad x * y = x.$$

$$\sigma(x * y) = \sigma(x) = \sigma(x) * \sigma(y) \quad (\forall \sigma \in \mathfrak{S}(E)),$$

$$|E| = 2 \notin \{0, 1, 3\}.$$

La même loi convient sur tout ensemble, notamment de cardinal 0, 1 ou 3.

Exercice 4

$$xy = 1, \quad yz = 1 \Rightarrow x = x(yz) = (xy)z = z \Rightarrow yx = yz = 1.$$

$$\forall x, \quad \exists y, \quad xy = yx = 1.$$

Exercice 5

1

$$x \circ y = yx, \quad (x \circ y) \circ z = z(yx) = (zy)x = x \circ (y \circ z).$$

$$1 \circ x = x \circ 1 = x.$$

2

$$M = \{1, a, b\}, \quad 1x = x1 = x, \quad ax = a, \quad bx = b \quad (x \in M).$$

$$x \neq 1 \Rightarrow (xy)z = xz = x = x(yz); \quad x = 1 \Rightarrow (xy)z = yz = x(yz).$$

$$\#\{z : \forall x, zx = z\} = 2, \quad \#\{z : \forall x, xz = z\} = 0.$$

Un isomorphisme avec le monoïde opposé échangerait ces deux ensembles : impossible.

3

$$x \circ x^{-1} = x^{-1}x = 1 = x^{-1} \circ x.$$

$$\varphi(x) = x^{-1}, \quad \varphi(xy) = y^{-1}x^{-1} = \varphi(x) \circ \varphi(y), \quad \varphi^2 = \text{id}.$$

Autocorrection B

(i)

$$\ell(uv) = \ell(u) + \ell(v), \quad \ell(u) > 0 \Rightarrow uv \neq \emptyset.$$

Pas un groupe.

(ii)

$$0M = 0 \neq I_2.$$

Pas un groupe.

(iii)

$$\det(AB) = \det A \det B, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad (ad - bc = 1).$$

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad UV = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = VU.$$

Groupe non abélien.

(iv)

$$U(x)U(y) = U(x+y), \quad U(0) = I_2, \quad U(x)^{-1} = U(-x).$$

Groupe abélien, isomorphe à $(\mathbb{R}, +)$.

(v)

$$e \in \mathbb{R}, \quad x = e - 1 \Rightarrow \max(e, x) = e \neq x.$$

Aucun neutre.

(vi)

$$z^n = w^m = 1 \Rightarrow (zw)^{nm} = 1, \quad (z^{-1})^n = 1.$$

Sous-groupe du groupe abélien $\mathbb{C}^\times$.

Exercice 6

$$z = xy \Rightarrow z^2 = xyxy = y^4 \Rightarrow zy^4 = y^4z.$$

$$y^5 = 1 \Rightarrow (y^4)^4 = y^{16} = y \Rightarrow zy = yz.$$

$$x = zy^{-1} \Rightarrow xy = yx, \quad x^2y = xyx = y^3 \Rightarrow x^2 = y^2.$$

Exercice 7

$$g^2 = 1 \Rightarrow g^{-1} = g, \quad xy = (xy)^{-1} = y^{-1}x^{-1} = yx.$$

Exercice 8

$$\mathbf{1}_{A \triangle B} = \mathbf{1}_A + \mathbf{1}_B \pmod{2}.$$

$$\mathbf{1}_{(A \triangle B) \triangle C} = \mathbf{1}_A + \mathbf{1}_B + \mathbf{1}_C = \mathbf{1}_{A \triangle (B \triangle C)}.$$

$$A \triangle \emptyset = A, \quad A \triangle A = \emptyset, \quad A \triangle B = B \triangle A.$$

Autocorrection C

$$H = \bigcup_{n \geq 0} H_n, \quad 1 \in H_0 \subset H.$$

$$x \in H_p, y \in H_q \Rightarrow x, y \in H_{\max(p,q)} \Rightarrow xy^{-1} \in H.$$

Autocorrection D

1(a)

$$1 \in C_G(g), \quad gx = xg, gy = yg \Rightarrow gxy^{-1} = xgy^{-1} = xy^{-1}g.$$

1(b)

$$h \in C_G(g) \iff gh = hg \iff \varphi(g)\varphi(h) = \varphi(h)\varphi(g).$$

$$\varphi \text{ bijective} \Rightarrow \varphi(C_G(g)) = C_G(\varphi(g)).$$

2(a)

$$Z(G) = \bigcap_{g \in G} C_G(g) \leq G.$$

2(b)

$$\varphi(Z(G)) = \bigcap_{g \in G} \varphi(C_G(g)) = \bigcap_{g \in G} C_G(\varphi(g)) = Z(G).$$

Exercice 9

1

$$0 \in \text{Pér}(f), \quad S, T \in \text{Pér}(f) \Rightarrow f(x + S - T) = f(x - T) = f(x).$$

2

$$f = \mathbf{1}_G, \quad T \in G \Rightarrow (x + T \in G \iff x \in G) \Rightarrow T \in \text{Pér}(f).$$

$$T \in \text{Pér}(f) \Rightarrow f(T) = f(0) = 1 \Rightarrow T \in G.$$

Exercice 10

1

$$H_1 \subseteq H_2 \Rightarrow H_1 \cup H_2 = H_2 \leq G.$$

$$a \in H_1 \setminus H_2, b \in H_2 \setminus H_1 : ab \in H_1 \Rightarrow b \in H_1, \quad ab \in H_2 \Rightarrow a \in H_2.$$

$$ab \notin H_1 \cup H_2 \Rightarrow H_1 \cup H_2 \text{ non stable.}$$

2

Si H_1, H_2 étaient tous deux stricts et $G = H_1 \cup H_2$, ils ne peuvent être inclus l'un dans l'autre, car alors leur union serait le plus grand, encore strict. Prenons donc $a \in H_1 \setminus H_2$ et $b \in H_2 \setminus H_1$. Alors ab n'appartient ni à H_1 (sinon $b = a^{-1}(ab) \in H_1$) ni à H_2 (sinon $a = (ab)b^{-1} \in H_2$), contradiction. Ainsi $G = H_1$ ou $G = H_2$.

3

$$\mathbb{F}_2^2 = \{0, (1, 0)\} \cup \{0, (0, 1)\} \cup \{0, (1, 1)\}.$$

Exercice 11

g d'ordre infini $\Rightarrow \langle g \rangle \supsetneq \langle g^2 \rangle \supsetneq \langle g^4 \rangle \supsetneq \dots$,

impossible. Tous les sous-groupes cycliques sont finis.

$$\{\langle g \rangle : g \in G\} = \{C_1, \dots, C_r\} \Rightarrow G = \bigcup_{i=1}^r C_i \text{ fini.}$$

Exercice 12

1

$$f(a) \circ f(b) = f(a * b), \quad (f(a) \circ f(b)) \circ f(c) = f((a * b) * c) = f(a * (b * c)) = f(a) \circ (f(b) \circ f(c)).$$

$$e_E = f(e_G), \quad f(a)^{-1} = f(a^{-1}).$$

2

$|E| = n \geq 1 \Rightarrow E \simeq \mathbb{Z}/n\mathbb{Z}$ par transport de structure.

$E = \emptyset \Rightarrow$ aucun neutre.

3

$$f(t) = \frac{e^{2t} - 1}{e^{2t} + 1}, \quad f^{-1}(x) = \frac{1}{2} \ln \frac{1+x}{1-x}.$$

$$f : \mathbb{R} \xrightarrow{\sim} (-1, 1), \quad f(s+t) = \frac{f(s) + f(t)}{1 + f(s)f(t)}.$$

$$e = 0, \quad x^{-1} = -x.$$

Exercice 13

1

$$(xy)^2 = x^2y^2 \iff xyxy = xxyy \iff yx = xy.$$

2

$$s(x) = x^2, \quad \ker s = \{x : x^2 = 1\}, \quad \text{ord}(x) \mid 2, |G| \Rightarrow x = 1.$$

s injective, G fini $\Rightarrow s$ surjective.

3

$$m = |G| \text{ impair}, \quad y^m = 1, \quad x = y^{(m+1)/2} \Rightarrow x^2 = y^{m+1} = y.$$

Exercice 14

$$\varphi(x_1, x_2)\varphi(y_1, y_2) = x_1x_2y_1y_2 = x_1y_1x_2y_2 = \varphi(x_1y_1, x_2y_2).$$

$$x_1x_2 = 1 \Rightarrow x_1 = x_2^{-1} \in H_1 \cap H_2 = \{1\} \Rightarrow \ker \varphi = \{(1, 1)\}.$$

$$\text{Im } \varphi \leq G, \quad H_1 \cup H_2 \subset \text{Im } \varphi \Rightarrow \text{Im } \varphi = G.$$

Exercice 15

$$(-1)^2 = 1, \quad -1 \neq 1; \quad 2x = 0 \Rightarrow x = 0 \text{ dans } \mathbb{R}.$$

Les éléments d'ordre 2 sont préservés par isomorphisme.

Exercice 16

$$\Phi(z) = \left(\frac{z}{|z|}, \ln |z| \right), \quad \Phi^{-1}(u, t) = ue^t.$$

$$\Phi(zw) = \left(\frac{z}{|z|} \frac{w}{|w|}, \ln |z| + \ln |w| \right) = \Phi(z)\Phi(w).$$

Exercice 17

1

$$n \geq 1, \quad \varphi : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z} \Rightarrow n\varphi(\bar{1}) = \varphi(\bar{0}) = 0 \Rightarrow \varphi = 0.$$

2

$$q \in \mathbb{Q}, \quad k \geq 1 : \quad \varphi(q) = k\varphi(q/k) \in k\mathbb{Z}.$$

$$\bigcap_{k \geq 1} k\mathbb{Z} = \{0\} \Rightarrow \varphi = 0.$$

Exercice 18

$$m = |F|, \quad \varphi(q) = \varphi(q/m)^m = 1_F.$$

Exercice 19

1

$$\varphi : \mathbb{Z}^n \rightarrow F \Rightarrow \varphi(k_1, \dots, k_n) = \varphi(e_1)^{k_1} \dots \varphi(e_n)^{k_n}.$$

$$\varphi \mapsto (\varphi(e_1), \dots, \varphi(e_n)) \text{ injective} \Rightarrow |\text{Hom}(\mathbb{Z}^n, F)| \leq |F|^n.$$

2

$$a \in \mathbb{F}_2^n, \quad \varphi_a(k) = \sum_{j=1}^n a_j k_j \pmod{2}.$$

$$|\text{Hom}(\mathbb{Z}^n, \mathbb{F}_2)| = 2^n.$$

3

$$\mathbb{Z}^n \simeq \mathbb{Z}^m \Rightarrow 2^n = 2^m \Rightarrow n = m.$$

La réciproque est immédiate.

Exercice 20

$$\sigma_g \sigma_h = \sigma_{gh}, \quad \sigma_g^{-1} = \sigma_{g^{-1}}.$$

$$\sigma_g = \sigma_h \Rightarrow g = \sigma_g(1) = \sigma_h(1) = h.$$

Autocorrection E

$$\zeta = e^{2i\pi/12}, \quad i = \zeta^3, \quad j = \zeta^4, \quad i^{-1}j = \zeta.$$

$$\langle i, j \rangle = \langle \zeta \rangle = \mathbb{U}_{12}.$$

Autocorrection F

$$1. \quad (x^{-1})^k = 1 \iff x^k = 1.$$

$$2. \quad (xyx^{-1})^k = yx^k y^{-1} = 1 \iff x^k = 1.$$

$$3. \quad yx = x^{-1}(xy)x \Rightarrow \text{ord}(yx) = \text{ord}(xy).$$

Exercice 21

1

$$G = \langle S \rangle, \quad |S| = r < \infty, \quad T = S \cup S^{-1}, \quad A_n = \{t_1 \cdots t_k : 0 \leq k \leq n, t_i \in T\}.$$

$$|A_n| \leq \sum_{k=0}^n (2r)^k < \infty, \quad A_n \subseteq A_{n+1}, \quad G = \bigcup_{n \geq 0} A_n.$$

2

$$q_i = a_i/b_i, \quad D = \prod_{i=1}^r b_i \neq 0 \Rightarrow \langle q_1, \dots, q_r \rangle \subseteq D^{-1}\mathbb{Z}.$$

$$1/(2D) \notin D^{-1}\mathbb{Z} \Rightarrow \mathbb{Q} \text{ n'est pas de type fini.}$$

Exercice 22

$$x \neq 1 \Rightarrow \langle x \rangle = G.$$

$$\text{ord}(x) = \infty \Rightarrow \{1\} \subsetneq \langle x^2 \rangle \subsetneq G, \quad \perp.$$

$$|G| = n, \quad n \text{ composé} \Rightarrow \exists p \text{ premier}, \quad 1 < p < n, \quad p \mid n.$$

$$|\langle x^p \rangle| = n/p \in \{2, \dots, n-1\}, \quad \perp.$$

$$G \simeq \mathbb{Z}/p\mathbb{Z} \quad (p \text{ premier}).$$

Exercice 23

$$M \subseteq G, \quad 1 \in M, \quad MM \subseteq M.$$

$$x \in M, \quad d = \text{ord}(x) < \infty \Rightarrow x^{-1} = x^{d-1} \in M.$$

Exercice 24

1

$$x \neq 1, \quad \text{ord}(x) \mid p \Rightarrow \text{ord}(x) = p \Rightarrow G = \langle x \rangle.$$

2

$$|G| = 4 : \quad \exists x, \text{ord}(x) = 4 \Rightarrow G \simeq C_4.$$

$$\text{Sinon, } x^2 = 1 \ (\forall x), \quad G \text{ abélien, } \quad a, b \neq 1, \quad a \neq b \Rightarrow G = \{1, a, b, ab\} \simeq C_2^2.$$

$$|G| = 6 : \quad \text{l'involution } x \mapsto x^{-1} \text{ possède un nombre pair de points fixes.}$$

$$1 \text{ est fixe} \Rightarrow \exists b \neq 1, \quad b^2 = 1.$$

$$\forall x \neq 1, \text{ord}(x) = 2 \Rightarrow G \text{ abélien} \Rightarrow \exists H \simeq C_2^2 \leq G, \quad 4 \mid 6, \quad \perp.$$

$$\exists a, \quad \text{ord}(a) \in \{3, 6\}.$$

$$\text{ord}(a) = 6 \Rightarrow G \simeq C_6.$$

$$\text{ord}(a) = 3 : \quad H = \langle a \rangle, \quad G = H \sqcup bH = H \sqcup Hb, \quad bab^{-1} \in \{a, a^{-1}\}.$$

$$bab^{-1} = a \Rightarrow G \text{ abélien.} \quad \text{ord}(b) \in \{2, 6\}.$$

$$\text{Si } \text{ord}(b) = 6, \text{ alors } G \simeq C_6; \text{ si } \text{ord}(b) = 2, \text{ alors } ab \text{ a ordre 6. Ainsi } G \simeq C_6.$$

$$bab^{-1} = a^{-1} \Rightarrow a^3 = b^2 = 1, \quad ba = a^{-1}b, \quad G = \{a^i b^j : 0 \leq i < 3, 0 \leq j < 2\} \simeq \mathfrak{S}_3.$$

Ordre	Groupes
1	$\{1\}$
2, 3, 5, 7	C_2, C_3, C_5, C_7 respectivement
4	$C_4, C_2 \times C_2$
6	$C_6, \mathfrak{S}_3$

Exercice 25

1

$$\text{ord}(g) \in \{1, 2, 4, 8\}.$$

$$\text{ord}(g) = 8 \Rightarrow G \text{ cyclique}; \quad \forall g, g^2 = 1 \Rightarrow G \text{ abélien}.$$

$$G \text{ non abélien} \Rightarrow \exists x, \quad \text{ord}(x) = 4.$$

2

$$H = \langle x \rangle, \quad y \notin H \Rightarrow H \cap yH = \emptyset, \quad |H| + |yH| = 8.$$

$$G = H \sqcup yH = \langle x, y \rangle, \quad xy = yx \Rightarrow G \text{ abélien}, \quad \perp.$$

3

$$yH = G \setminus H = Hy \Rightarrow yHy^{-1} = H.$$

$$\text{ord}(yxy^{-1}) = 4 \Rightarrow yxy^{-1} \in \{x, x^{-1}\} \Rightarrow yxy^{-1} = x^{-1}.$$

4

$$y^2 \in H, \quad y^2 = x^k, \quad yx^ky^{-1} = y^2 = x^k = x^{-k} \Rightarrow k \in \{0, 2\} \pmod{4}.$$

$$x^4 = 1, \quad yx = x^{-1}y, \quad y^2 = 1 \text{ ou } x^2.$$

$$G = \{x^i y^j : 0 \leq i < 4, j \in \{0, 1\}\}.$$

Ces relations déterminent tous les produits :

$$y^2 = 1 \Rightarrow G \simeq D_8, \quad y^2 = x^2 \Rightarrow G \simeq Q_8.$$

$$D_8 : 5 \text{ éléments d'ordre } 2; \quad Q_8 : 1 \text{ élément d'ordre } 2.$$

Exercice 26

1

$$g \notin H \Rightarrow gH = Hg = G \setminus H.$$

$$g, h \notin H \Rightarrow h \in g^{-1}H \Rightarrow gh \in H.$$

$$g \in H, h \notin H \Rightarrow gh \notin H; \quad g, h \in H \Rightarrow gh \in H.$$

$$\varphi(gh) = \varphi(g)\varphi(h).$$

2

$$\chi_a(x) = (-1)^{a_1x_1 + \dots + a_dx_d}, \quad a \in \mathbb{F}_2^d \setminus \{0\}.$$

$$\chi_a \text{ surjective} \Rightarrow |\ker \chi_a| = 2^{d-1}.$$

$$H \text{ d'indice } 2 \Rightarrow \chi_H(x) = \begin{cases} 1 & x \in H, \\ -1 & x \notin H \end{cases} \text{ unique.}$$

$$2^d - 1 \text{ sous-groupes} \quad (d \geq 1).$$

3

$$H_1 \neq H_2 \Rightarrow \chi_1 \neq \chi_2, \quad \chi_3 = \chi_1\chi_2.$$

$$\chi_3 \neq 1, \quad \chi_3 \neq \chi_1, \chi_2 \Rightarrow \ker \chi_3 \text{ troisième sous-groupe d'indice } 2.$$

Exercice 27

1

$$h \mapsto (h, h^{-1}g) : G \xrightarrow{\sim} \{(h, k) : hk = g\}.$$

2

$$m = |G|, \quad M^2 = \frac{1}{m^2} \sum_{h,k \in G} hk = \frac{1}{m^2} \sum_{g \in G} mg = M.$$

Exercice 28

L'assertion est fausse pour des matrices.

$$T = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad G = \{I_2, T\}, \quad r = 1, \quad \prod_{g \in G} g = T \neq -I_2.$$

La formule valable est

$$\prod_{g \in G} g = \prod_{\substack{g \in G \\ g^2 = I_n}} g,$$

les autres facteurs se regroupant par paires g, g^{-1} .

Exercice 29

1

$$i^2(x) = (x^{-1})^{-1} = x, \quad \text{Fix}(i) = \{x : x^2 = 1\}.$$

2

$$|G| = |\text{Fix}(i)| + 2r \text{ pair} \Rightarrow |\text{Fix}(i)| \text{ pair}.$$

$$1 \in \text{Fix}(i) \Rightarrow \exists x \neq 1, \quad x^2 = 1 \Rightarrow \text{ord}(x) = 2.$$

Exercice 30

1

$$C_n = \langle \bar{1} \rangle \neq \{0\} \Rightarrow \text{rg}(C_n) = 1.$$

$$C_n^2 = \langle (1,0), (0,1) \rangle, \quad \text{ord}(x) \leq n < n^2 \Rightarrow \text{rg}(C_n^2) = 2.$$

$$\mathfrak{S}_3 = \langle (12), (23) \rangle \text{ non abélien} \Rightarrow \text{rg}(\mathfrak{S}_3) = 2.$$

2(a)

$$G_1 = \langle S_1 \rangle, \quad G_2 = \langle S_2 \rangle \Rightarrow G_1 \times G_2 = \langle (S_1 \times \{1\}) \cup (\{1\} \times S_2) \rangle.$$

$$\text{rg}(G_1 \times G_2) \leq \text{rg}(G_1) + \text{rg}(G_2).$$

2(b)

$$\text{ord}(1,1) = 6 \text{ dans } C_2 \times C_3 \Rightarrow \text{rg}(C_2 \times C_3) = 1 < 2.$$

3

$$H_0 = \{1\}, \quad H_k \neq G \Rightarrow g_{k+1} \notin H_k, \quad H_{k+1} = \langle H_k, g_{k+1} \rangle.$$

$$H_k \cap g_{k+1}H_k = \emptyset \Rightarrow |H_{k+1}| \geq 2|H_k|.$$

$$H_m = G \Rightarrow 2^m \leq |G|, \quad G = \langle g_1, \dots, g_m \rangle \Rightarrow \text{rg}(G) \leq \log_2 |G|.$$

4

$$G = \mathbb{F}_2^d, \quad |G| = 2^d, \quad \langle r \text{ éléments} \rangle \text{ a au plus } 2^r \text{ éléments}.$$

$$\text{rg}(G) \geq d, \quad G = \langle e_1, \dots, e_d \rangle \Rightarrow \text{rg}(G) = d.$$

Exercice 31

1

$$g \in G, \quad |A| + |gB^{-1}| = |A| + |B| > |G| \Rightarrow A \cap gB^{-1} \neq \emptyset.$$

$$a = gb^{-1} \Rightarrow g = ab \Rightarrow AB = G.$$

2

$$G = C_2, \quad A = B = \{0\}, \quad |A| + |B| = |G|, \quad A + B = \{0\} \neq G.$$

Exercice 32

Dans les deux formules imprimées, lire $xy = yx$. La probabilité est le nombre de couples divisé par $|G|^2$.

1

$$G/Z(G) = \langle aZ(G) \rangle \Rightarrow x = a^r z, \quad y = a^s w \quad (z, w \in Z(G)) \Rightarrow xy = a^{r+s} zw = yx.$$

$$G \text{ non abélien} \Rightarrow G/Z(G) \text{ non cyclique} \Rightarrow [G : Z(G)] \geq 4.$$

2

$$x \notin Z(G) \Rightarrow C_G(x) \subsetneq G \Rightarrow |C_G(x)| \leq |G|/2.$$

$$\frac{\#\{(x, y) : xy = yx\}}{|G|^2} = \frac{1}{|G|^2} \sum_{x \in G} |C_G(x)|$$

$$\leq \frac{|Z(G)|}{|G|} + \frac{|G| - |Z(G)|}{2|G|} = \frac{1}{2} + \frac{|Z(G)|}{2|G|} \leq \frac{5}{8}.$$

Exercice 33

$$\text{Lire } xHx^{-1} = \{xhx^{-1} : h \in H\}.$$

1

$$x \sim y \iff xHx^{-1} = yHy^{-1}.$$

L'égalité donne réflexivité, symétrie et transitivité.

$$(xh)H(xh)^{-1} = xHx^{-1} \quad (h \in H).$$

$$G/H \twoheadrightarrow \{xHx^{-1} : x \in G\} \Rightarrow r \leq |G|/|H|.$$

2

$$H_1, \dots, H_r \text{ conjugués distincts,} \quad 1 \in \bigcap_i H_i, \quad |H_i| = |H|.$$

$$\left| \bigcup_i H_i \right| \leq 1 + r(|H| - 1) \leq 1 + |G| - \frac{|G|}{|H|} < |G|.$$

Exercice 34

1

$$f_{a,b}(z) = az + b, \quad a \neq 0, \quad f_{a,b} \circ f_{c,d} = f_{ac, ad+b}.$$

$$\text{id} = f_{1,0}, \quad f_{a,b}^{-1} = f_{a^{-1}, -b/a}.$$

2

$$f_{u,v} f_{a,b} f_{u,v}^{-1} = f_{a, ub + (1-a)v}.$$

$$a \neq 1 \Rightarrow \{\text{conjugués de } f_{a,b}\} = \{f_{a,c} : c \in \mathbb{C}\}.$$

$$a = 1, b = 0 : \quad \{\text{id}\}; \quad a = 1, b \neq 0 : \quad \{f_{1,c} : c \in \mathbb{C}^*\}.$$

Exercice 35

1

$$I^2 = J^2 = -1, \quad IJ = K = -JI, \quad K^2 = -1.$$

$$JK = I = -KJ, \quad KI = J = -IK.$$

$$Q = \{\pm 1, \pm I, \pm J, \pm K\} \text{ stable par produit et inverse,} \quad Q = \langle I, J \rangle.$$

2

	1	-1	I	-I	J	-J	K	-K
1	1	-1	I	-I	J	-J	K	-K
-1	-1	1	-I	I	-J	J	-K	K
I	I	-I	-1	1	K	-K	-J	J
-I	-I	I	1	-1	-K	K	J	-J
J	J	-J	-K	K	-1	1	I	-I
-J	-J	J	K	-K	1	-1	-I	I
K	K	-K	J	-J	-I	I	-1	1
-K	-K	K	-J	J	I	-I	1	-1

3

$$\text{ord}(1) = 1, \quad \text{ord}(-1) = 2, \quad \text{ord}(\pm I) = \text{ord}(\pm J) = \text{ord}(\pm K) = 4.$$

$$Z(Q_8) = \{1, -1\}, \quad JIJ^{-1} = -I, \quad IJI^{-1} = -J, \quad IKI^{-1} = -K.$$

$$\{1\}, \{-1\}, \{I, -I\}, \{J, -J\}, \{K, -K\}.$$

Exercice 36

1

$$\sigma^2 = \tau^2 = \text{id}, \quad \gamma(x) = \tau(-x) = x + 1, \quad \gamma^n(x) = x + n.$$

$$\sigma^{-1}\gamma^n\sigma(x) = x - n = \gamma^{-n}(x).$$

2

$$\tau = \gamma\sigma, \quad \sigma\gamma^n = \gamma^{-n}\sigma, \quad \sigma^2 = 1.$$

Tout mot se réduit à γ^n ou $\sigma\gamma^n$; réciproquement, ces applications sont des mots en σ, τ .

3 et 4

$$T_n(x) = x + n, \quad R_n(x) = -x + n, \quad G = \{T_n, R_n : n \in \mathbb{Z}\}.$$

$$T_n T_m = T_{n+m}, \quad T_n R_m = R_{n+m}, \quad R_n T_m = R_{n-m}, \quad R_n R_m = T_{n-m}.$$

$$T_0 = 1, \quad T_n^{-1} = T_{-n}, \quad R_n^{-1} = R_n.$$

5

$$R_n T_1 = R_{n-1} \neq R_{n+1} = T_1 R_n.$$

$$T_n R_0 = R_n, \quad R_0 T_n = R_{-n} \Rightarrow n = 0.$$

$$Z(G) = \{T_0\}.$$

6

$$\varepsilon, \delta \in \{-1, 1\}, \quad \chi(T_n) = \varepsilon^n, \quad \chi(R_n) = \varepsilon^n \delta.$$

Les quatre règles de multiplication donnent $\chi(gh) = \chi(g)\chi(h)$.

$$\chi \mapsto (\chi(T_1), \chi(R_0)) \Rightarrow 4 \text{ morphismes.}$$

7

$$n \neq 0 \Rightarrow T_n^k = T_{kn} \neq 1 \quad (k \geq 1), \quad R_n^2 = 1.$$

$$T_k R_n T_{-k} = R_{n+2k}.$$

$$R_{2m} \sim R_0 = \sigma, \quad R_{2m+1} \sim R_1 = \tau.$$

Exercice 37

1

f isométrie $\Rightarrow f$ injective $\Rightarrow f$ bijective sur $\mathbb{U}_n$.

$$|f^{-1}(u) - f^{-1}(v)| = |u - v|.$$

Les compositions et l'identité préservent aussi les distances.

2

$$\zeta = e^{2i\pi/n}, \quad r^k(z) = \zeta^k z, \quad r^k = 1 \iff n \mid k \Rightarrow \text{ord}(r) = n.$$

3

$$f(1) = 1, \quad f(\zeta) = \zeta \Rightarrow |f(z) - 1| = |z - 1|, \quad |f(z) - \zeta| = |z - \zeta|.$$

$$|z| = |f(z)| = 1 \Rightarrow \text{Re } f(z) = \text{Re } z, \quad \text{Re}(f(z)\bar{\zeta}) = \text{Re}(z\bar{\zeta}).$$

$$\sin(2\pi/n) \neq 0 \Rightarrow \text{Im } f(z) = \text{Im } z \Rightarrow f(z) = z.$$

4

$$f(1) = \zeta^k, \quad g = r^{-k}f \Rightarrow g(1) = 1.$$

$$\min_{z \in \mathbb{U}_n \setminus \{1\}} |z - 1| = |\zeta - 1|, \quad \text{atteint en } \zeta, \zeta^{-1} \Rightarrow g(\zeta) \in \{\zeta, \zeta^{-1}\}.$$

$$c(z) = \bar{z} : \quad g(\zeta) = \zeta \Rightarrow g = 1; \quad g(\zeta) = \zeta^{-1} \Rightarrow cg = 1.$$

$$D_{2n} = \{r^k, r^k c : 0 \leq k < n\}, \quad |D_{2n}| = 2n.$$

5

$$crc = r^{-1} \neq r \quad (n \geq 3).$$

6

$$(r^k c)^2 = r^k r^{-k} = 1, \quad r^k c \neq 1.$$

$$\text{ord}(r^k) = 2 \iff n \text{ pair et } k = n/2 \pmod{n}.$$

$$r = (rc)c.$$

7

$$\chi(r) = \varepsilon, \quad \chi(c) = \delta, \quad \varepsilon, \delta \in \{-1, 1\}, \quad \varepsilon^n = 1.$$

$$\chi(r^k c^b) = \varepsilon^k \delta^b, \quad crc = r^{-1} \Rightarrow \delta \varepsilon \delta = \varepsilon^{-1}.$$

n impair : 2 morphismes; n pair : 4 morphismes.

Exercice 38

1

$$f_{a,b}(x) = ax + b, \quad a \in \mathbb{R}^*, \quad b \in \mathbb{R}.$$

$$f_{a,b} f_{c,d} = f_{ac, ad+b}, \quad f_{a,b}^{-1} = f_{a^{-1}, -b/a}, \quad 1 = f_{1,0}.$$

2

$$f, g \in \Gamma \Rightarrow fgf^{-1}g^{-1} \text{ a pour coefficient directeur } 1.$$

Une translation possède un point fixe $\iff$ elle est l'identité $\Rightarrow fg = gf$.

$\Gamma = \{1\} \Rightarrow$ tout ω convient.

$f_{a,b} \in \Gamma \setminus \{1\} \Rightarrow a \neq 1, \quad \omega = \frac{b}{1-a}$ unique point fixe de f .

$g \in \Gamma \Rightarrow f(g(\omega)) = g(f(\omega)) = g(\omega) \Rightarrow g(\omega) = \omega$.